

BOOK A DOCTOR SYSTEM

Project Features and Functional Modules

The Book a Doctor System is a full-stack MERN application that connects patients, doctors, and administrators through a secure appointment-booking platform. It supports role-based dashboards, doctor and patient management, appointment booking, profile management, and appointment-status tracking.

Core Functional Features

1. User Registration and Login Users can create accounts and log in using their email and password. Passwords are hashed with bcrypt before being stored, and successful login generates a JWT token.	2. Role-Based Access Control The system supports three roles: Admin, Doctor, and Patient. After login, each user is redirected to the correct dashboard and receives access only to the functions permitted for that role.
3. Admin Dashboard The administrator can monitor overall system activity through dashboard statistics, including total users, doctors, patients, appointments, and appointment-status counts.	4. Doctor Management The admin can view all doctors, add a new doctor, edit doctor details, and delete doctor records. Doctor information includes specialization, qualification, experience, hospital, consultation fee, address, availability, and profile details.
5. Patient Management The admin can view registered patients and delete patient records when required. Patient information is retrieved securely from MongoDB without exposing passwords.	6. Appointment Management Appointments can be viewed and managed by the appropriate roles. Appointment records contain the patient, doctor, date, time, reason, and current status.
7. Patient Doctor Browsing Patients can view available doctors along with specialization, hospital, experience, consultation fee, and other profile information.	8. Appointment Booking A patient can select a doctor, choose an appointment date and time, enter the reason for the visit, and submit the booking. The appointment is stored in MongoDB.
9. My Appointments Patients can view their booked appointments, doctor details, appointment date and time, reason, and live appointment status.	10. Doctor Appointment Dashboard Doctors can view only the appointments assigned to them, including patient name, phone, date, time, and reason.
11. Appointment Status Updates Doctors and authorized administrators can update an appointment to Pending, Confirmed, Cancelled, or Completed. The updated status is visible to the patient.	12. Profile Management Logged-in users can view and update their first name, last name, and phone number. Email is displayed as a protected account identifier.

Technical and Quality Features

Feature	Description
REST API Integration	The React frontend communicates with the Express backend through Axios using JSON-based REST APIs for authentication, doctors, users, appointments, dashboard statistics, and profiles.
MVC with Service Layer	The backend separates routes, middleware, controllers, services, models, and configuration. This makes the code modular, maintainable, and easier to test.
JWT Authentication	Protected API routes validate bearer tokens before allowing access to user-specific or administrator-only operations.
Password Security	bcryptjs is used to hash passwords before storage and compare passwords securely during login.
MongoDB Relationships	Appointments reference both patient and doctor user records using MongoDB ObjectId fields and Mongoose population.
Input Validation	Mongoose schemas define required fields and permitted appointment-status values to protect data integrity.
Security Middleware	CORS enables controlled frontend-backend communication, Helmet adds secure HTTP headers, and Morgan records API requests during development.
Responsive User Interface	React pages use reusable components, cards, tables, forms, status badges, and responsive CSS layouts for Admin, Doctor, and Patient modules.
API Testing	Postman and command-line requests are used to verify GET, POST, PUT, and DELETE endpoints.
Version Control	The complete client and server source code is maintained using Git and published in a GitHub repository.

Role-Based Feature Summary

Admin	View dashboard statistics; add, edit, and delete doctors; view and delete patients; view all appointments; update appointment status; manage profile.
Doctor	Log in securely; view assigned appointments and patient details; confirm, cancel, or complete appointments; manage profile.
Patient	Register and log in; browse doctors; book appointments; view personal appointment history and status; manage profile.

Project Outcome: The application provides a complete end-to-end doctor appointment workflow. Patients can book appointments, doctors can manage assigned appointments, and administrators can oversee users and platform operations. The MERN stack, JWT security, MVC structure, and REST API integration make the project suitable for academic demonstration and future enhancement.